Practical no 4 : Working with AWS RDS (Relational Database Service)

Aim : To create and use an AWS RDS database instance and perform basic database operations using a MySQL engine.

Tools / Requirements

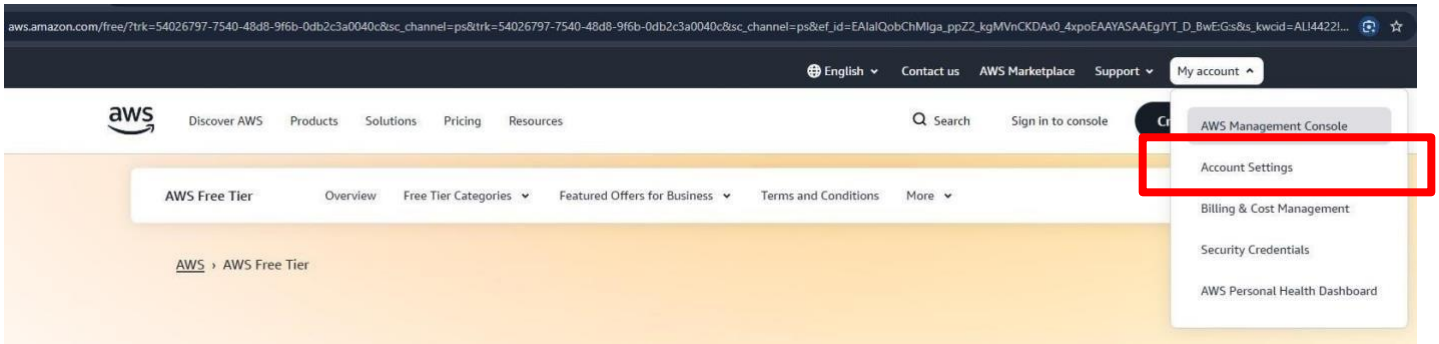
- AWS Account (Free Tier)
- Internet Browser
- MySQL Client (MySQL Workbench / phpMyAdmin / CLI)
- Basic knowledge of SQL

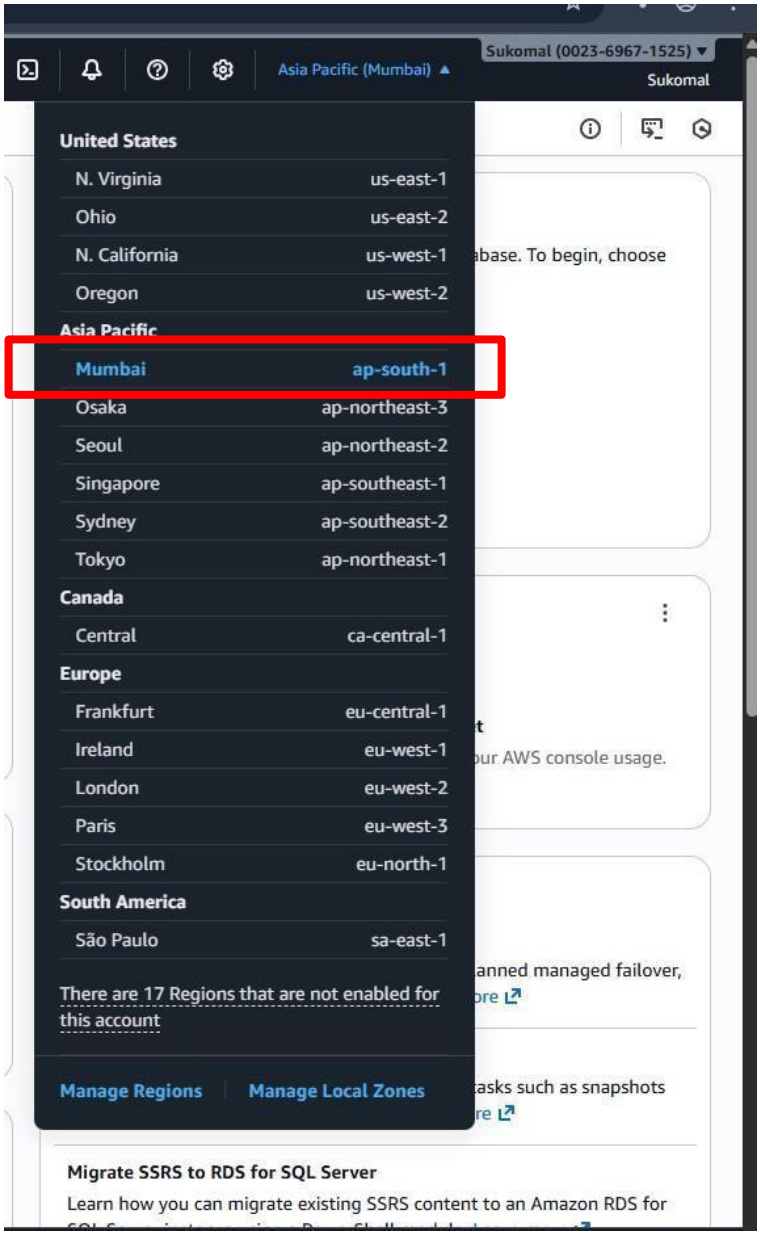
AWS RDS is a managed database service that supports relational databases like MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server. It handles: Database provisioning, Backups, Patching, ScalingHigh availability

Steps:

Step 1: Login to AWS Console

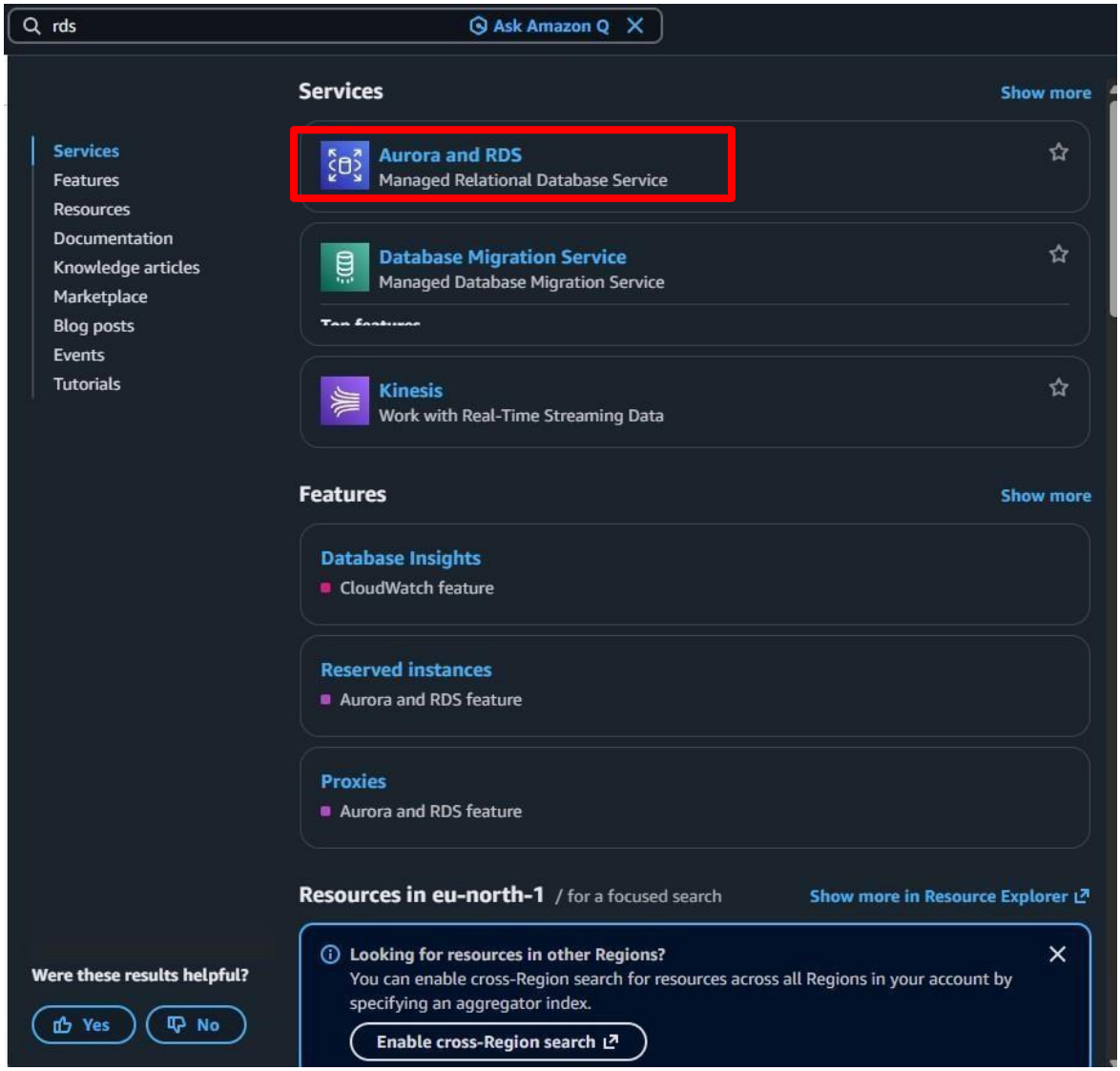
Go to https://aws.amazon.com
Login to AWS Management Console
Select a region (e.g., ap-south-1 (Mumbai))



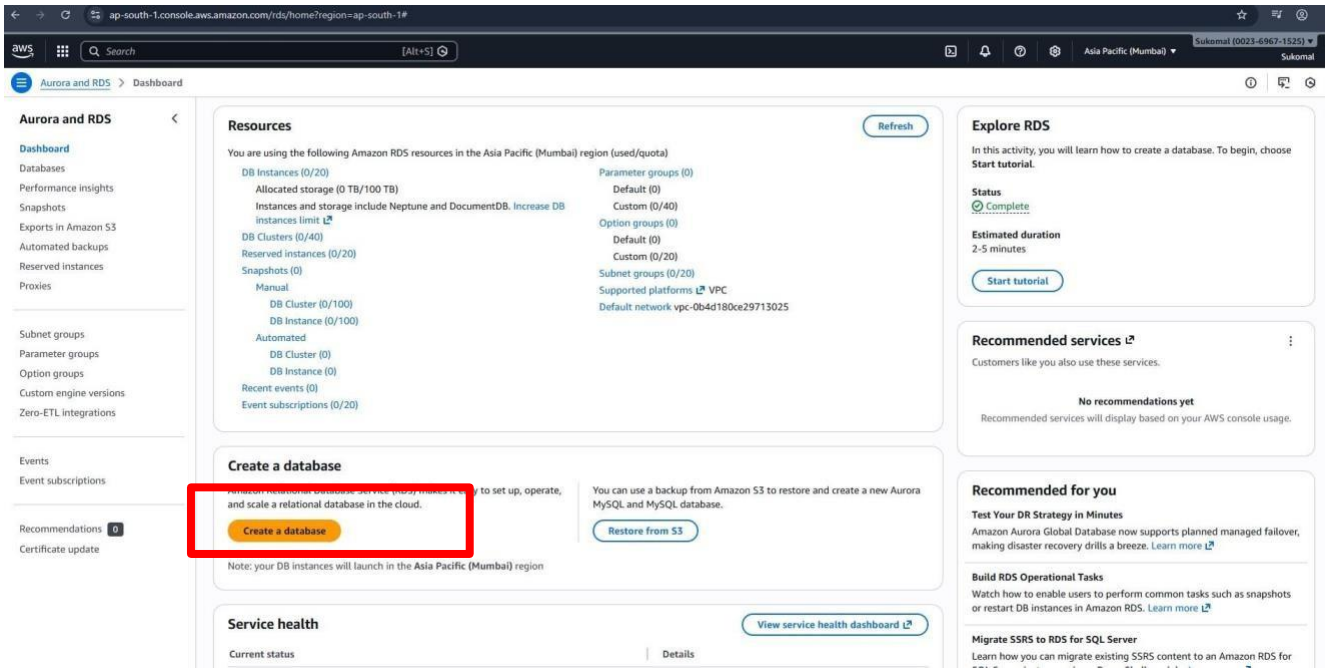


Step 2: Create an RDS Instance

Open Services → RDS



Click Create database

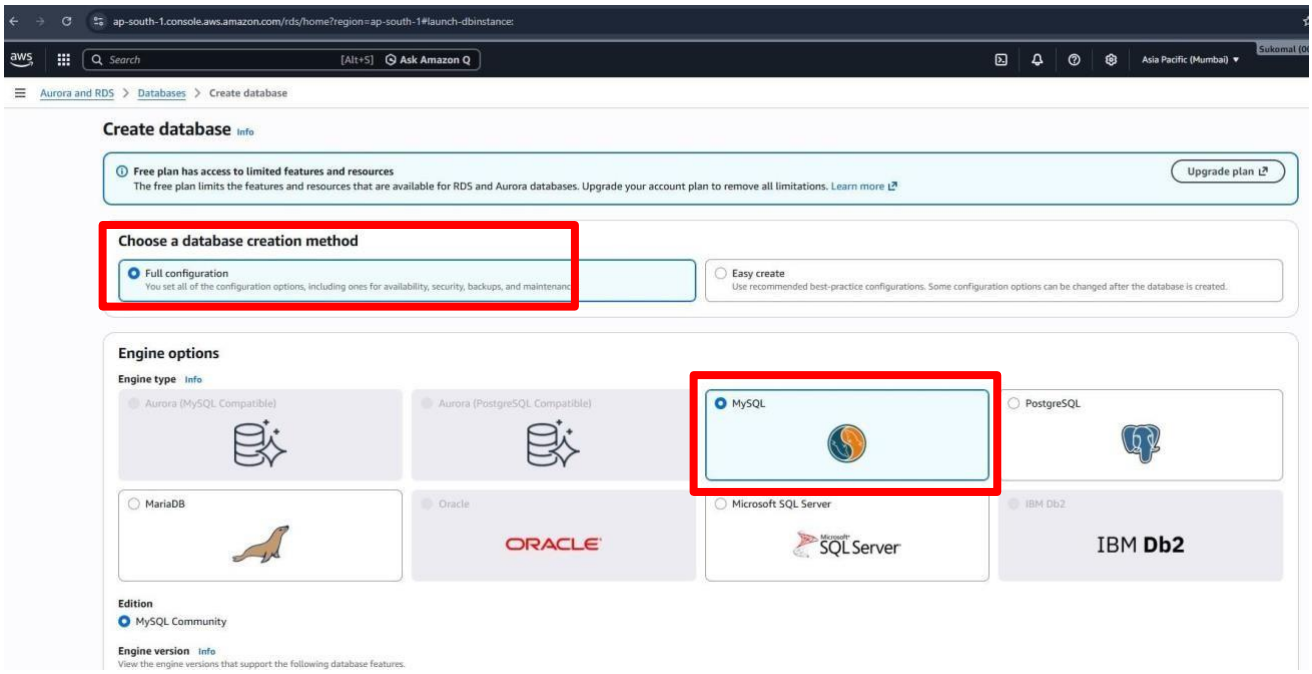


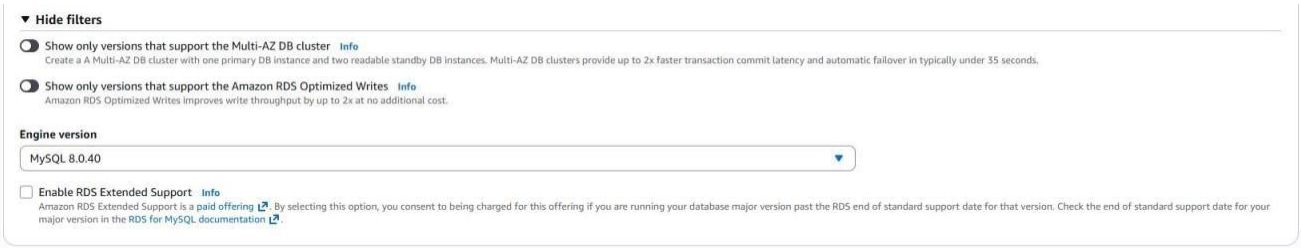
Choose:

Creation method: Standard create

Engine type: MySQL
Version: Default Templates:

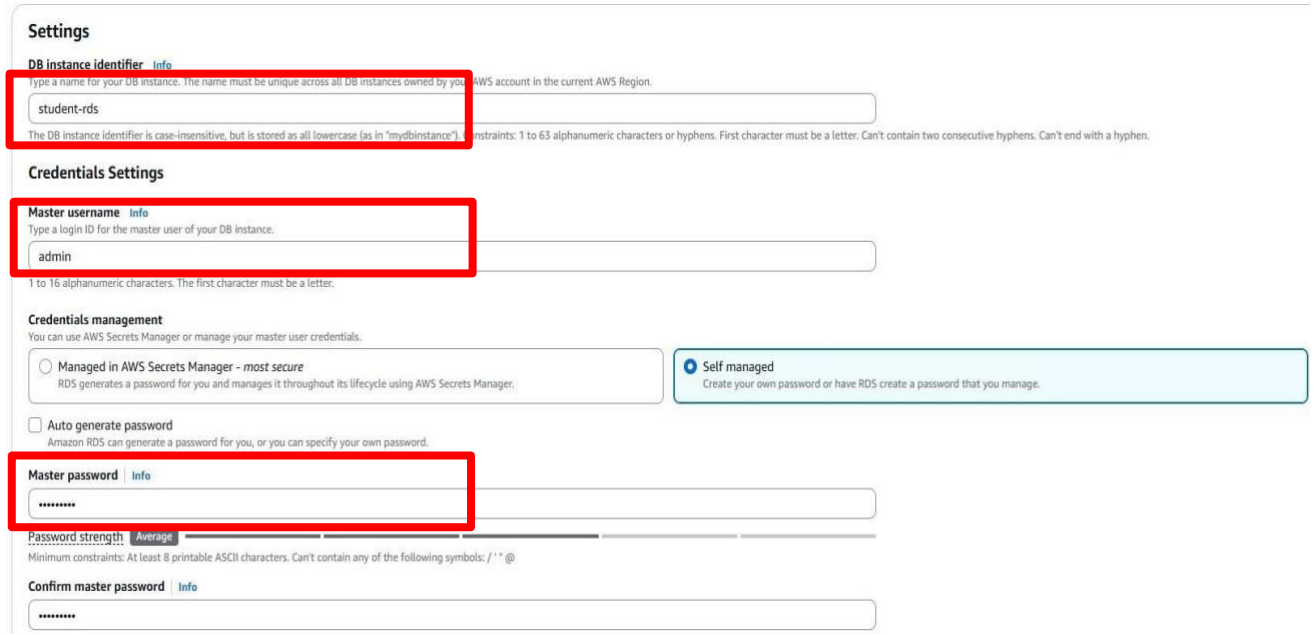
Select Free Tier

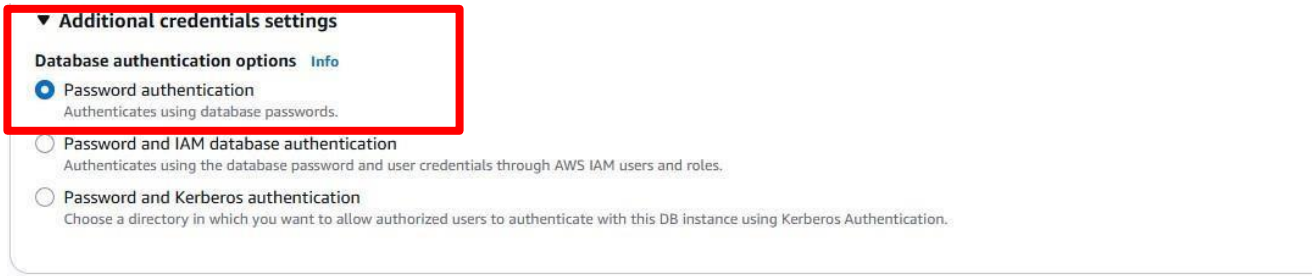


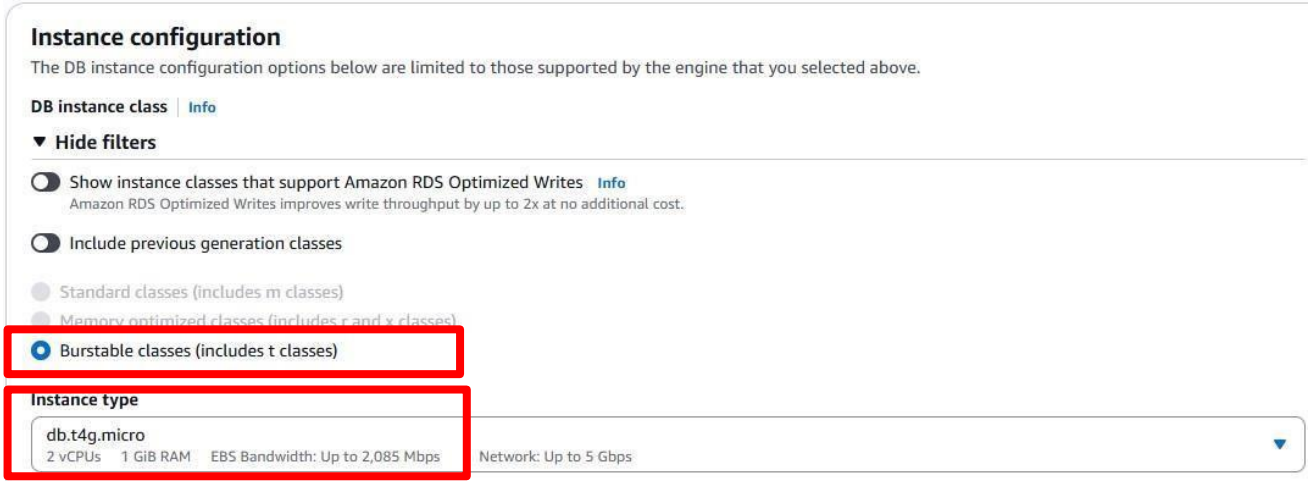


Step 3: Configure Database Settings DB instance identifier: student-rds Master username: admin

Password: TYCS_01
Instance class: db.t2.micro
Storage: Default (20 GB)







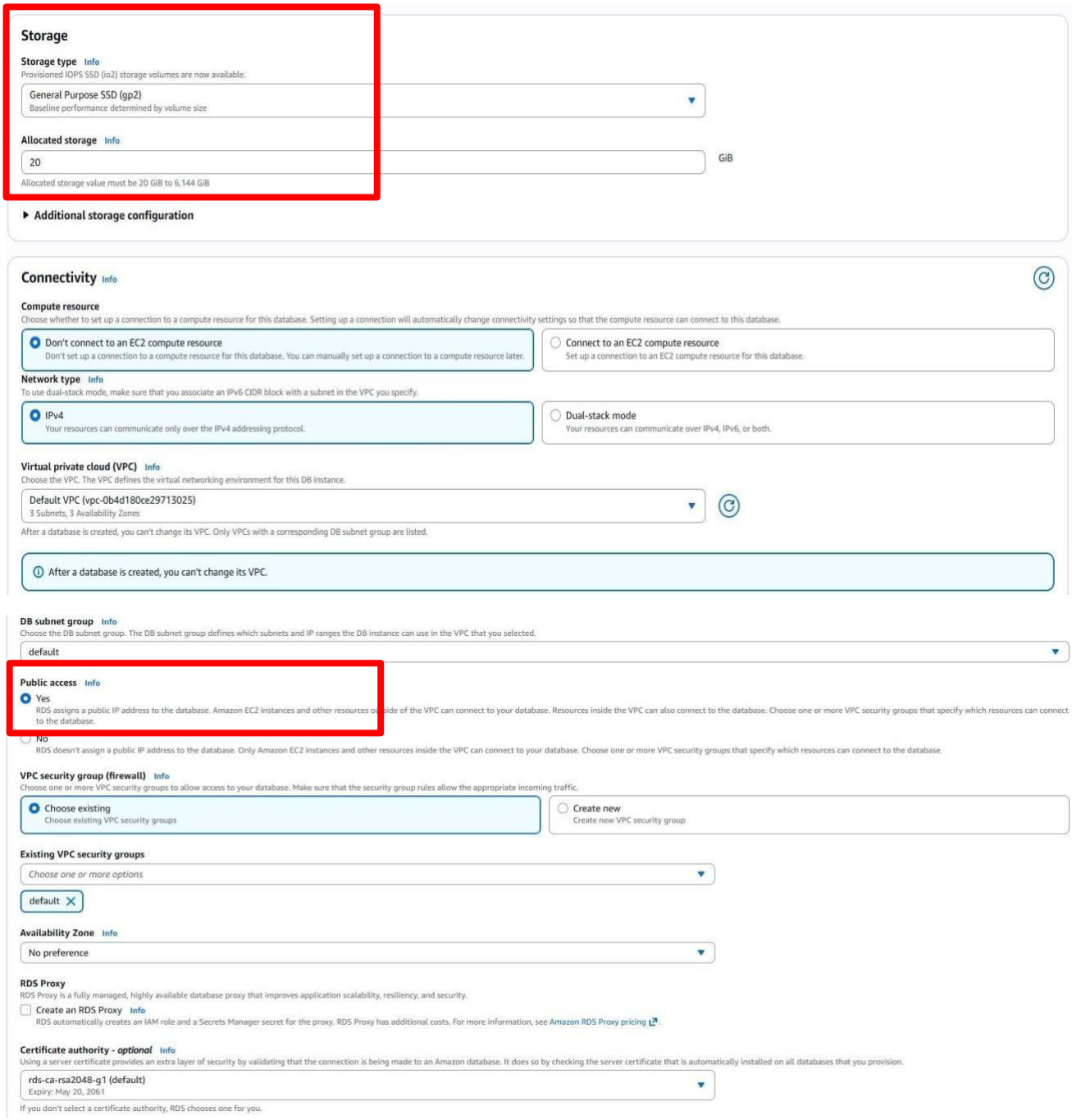
Step 4: Connectivity Settings

Public access: Yes

VPC Security Group:

Allow inbound rule for MySQL (port 3306)

Source: My IP


Storage
Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size
Allocated storage Info
20
GiB
Allocated storage value must be 20 GiB to 6,144 GiB
Additional storage configuration
Connectivity Info
Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
Network type Info
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.
IPv4
Your resources can communicate only over the IPv4 addressing protocol.
Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.
Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
Default VPC (vpc-0b4d180ce29713025)
3 Subnets, 3 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.
DB subnet group Info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.
default
Public access Info
Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.
VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group
Existing VPC security groups
Choose one or more options
default
Availability Zone Info
No preference
RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
Create an RDS Proxy Info
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.
Certificate authority - optional Info
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061
If you don't select a certificate authority, RDS chooses one for you.

▼ Additional configuration

Database port Info

TCP/IP port that the database will use for application connections.

3306

Tags - *optional*

A tag consists of a case-sensitive key-value pair.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Monitoring Info

Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. **Database Insights** pricing is separate from RDS monthly estimates. See Amazon CloudWatch pricing.

Database Insights - Advanced

- Retains 15 months of performance history
- Fleet-level monitoring
- Integration with CloudWatch Application Signals

Database Insights - Standard

▼ **Additional monitoring settings**

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

☐ Audit log

▼ **Additional monitoring settings**

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

☐ Audit log

☐ Error log

☐ General log

☐ iam-db-auth-error log

☐ Slow query log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

▼ Additional configuration

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name Info

test_mysql

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group Info

default.mysql8.0

Option group Info

default:mysql-8-0

Backup

☑ Enable automated backup

Creates a point-in-time snapshot of your database

⚠ Please note that automated backups are currently supported for InnoDB storage engine only. If you are using MyISAM, refer to details here.

Backup retention period Info

The number of days (1-35) for which automatic backups are kept.

1 day

Backup window Info

The daily time range (in UTC) during which RDS takes automated backups.

○ Choose a window

◉ No preference

Backup tags | Info

Copies tags from the source database to the automated backup and snapshots respectively.

- [ ] Copy tags to automated backup
  This is a one-time setting. Future tag modifications need manual updates.
- [x] Copy tags to snapshots

Backup replication Info

- [ ] Enable replication in another AWS Region
  Enabling replication automatically creates backups of your DB instance in the selected Region, for disaster recovery, in addition to the current Region.

- [x] Enable encryption
  Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. Info

AWS KMS key Info

(default) aws/rds

Account

002369671525

KMS key ID

alias/aws/rds

Maintenance

Auto minor version upgrade Info

- [x] Enable auto minor version upgrade
  Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version documentation.

Maintenance window Info

Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

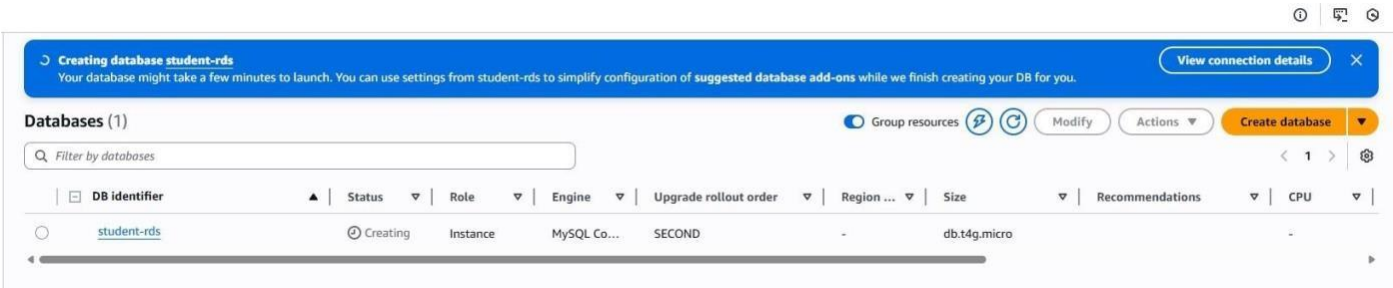
- ( ) Choose a window
- (•) No preference

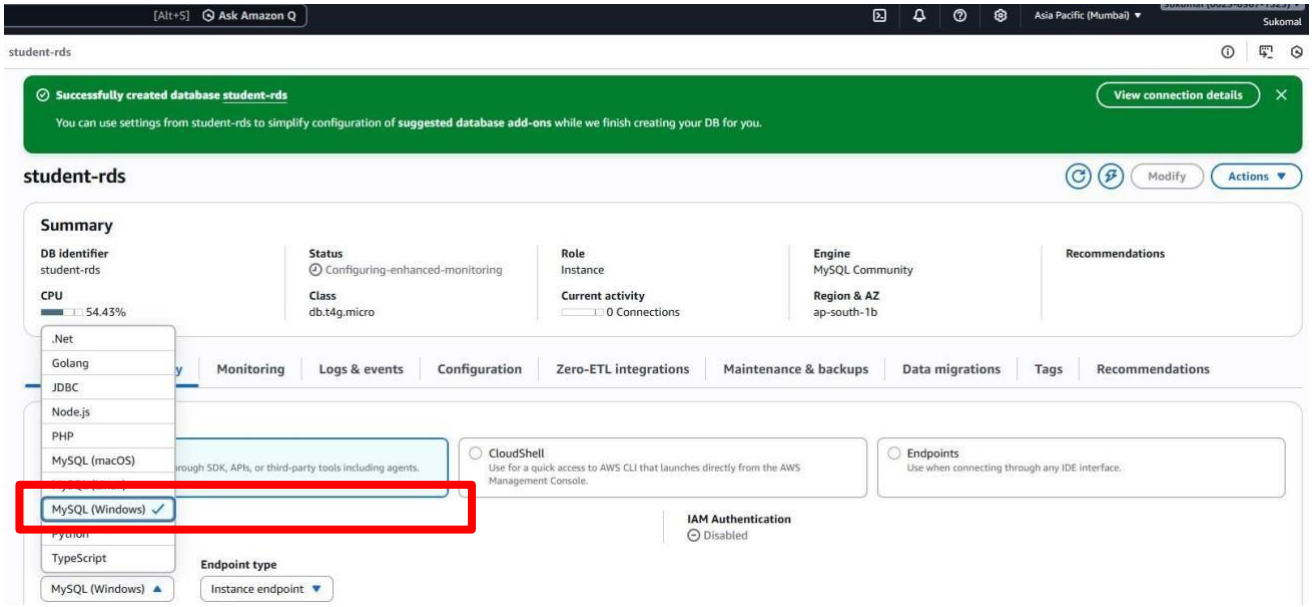
- [ ] Enable deletion protection
  Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Step 5: Create Database

Click Create database

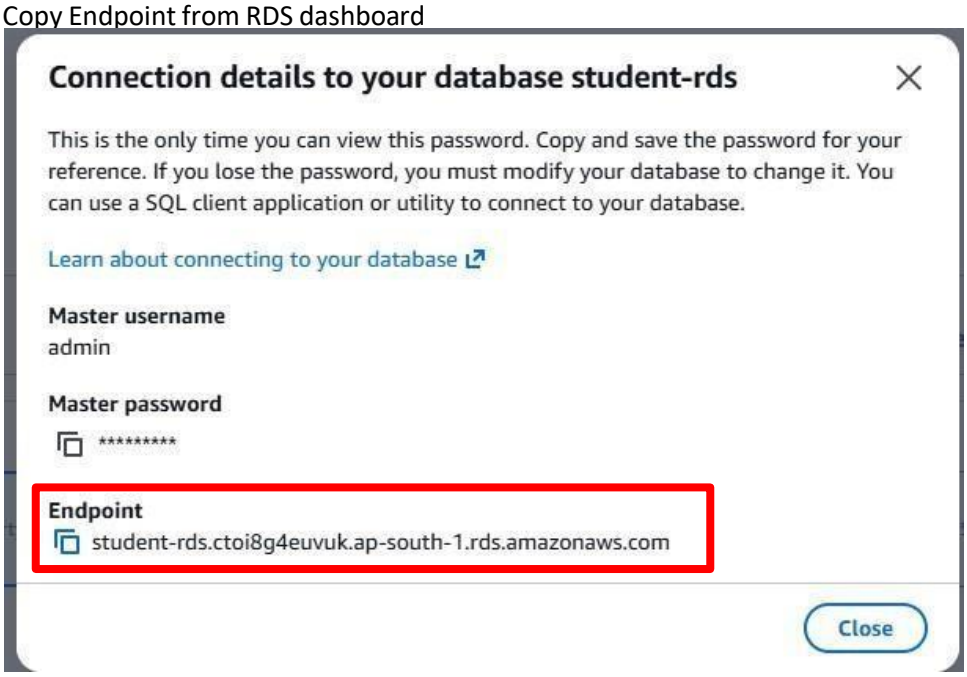
Wait for status to become Available





Step 6: Connect to RDS Database

Copy Endpoint from RDS dashboard



Open MySQL client

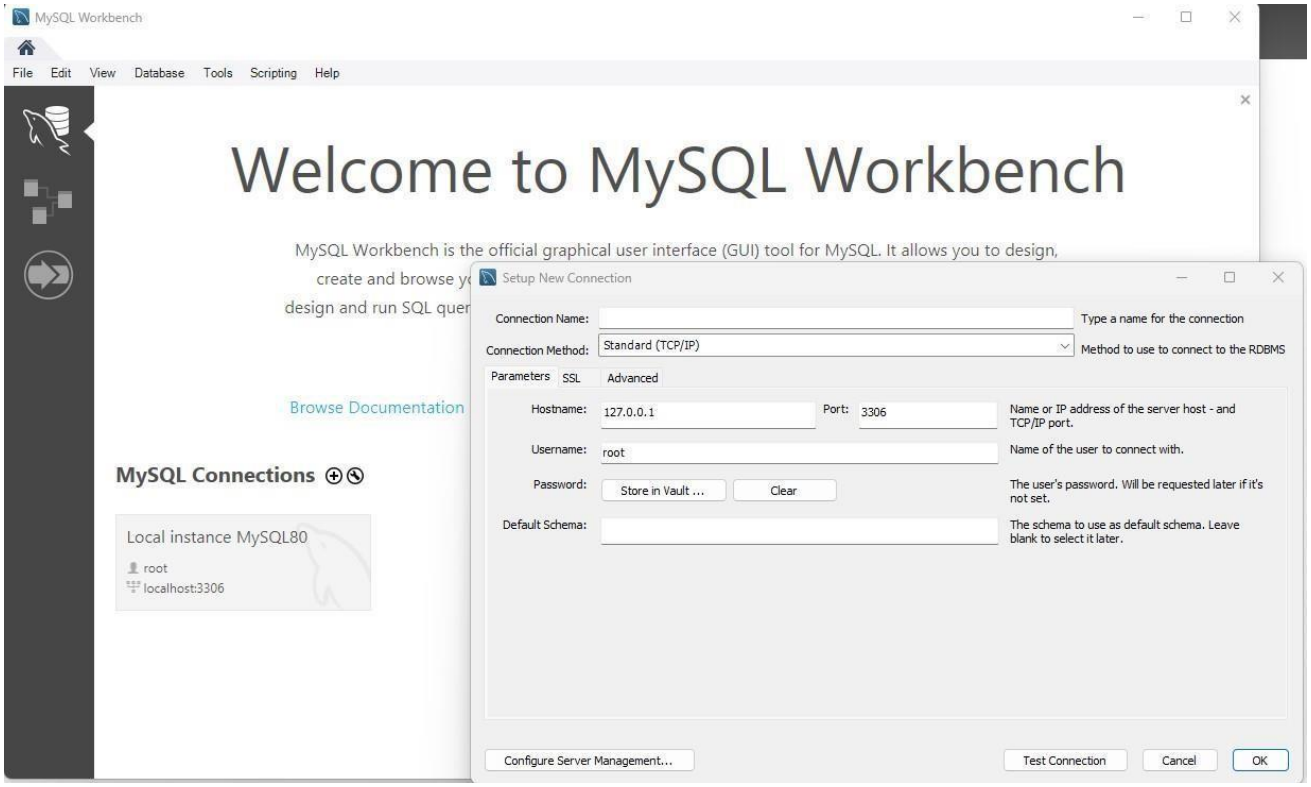
Connect using:

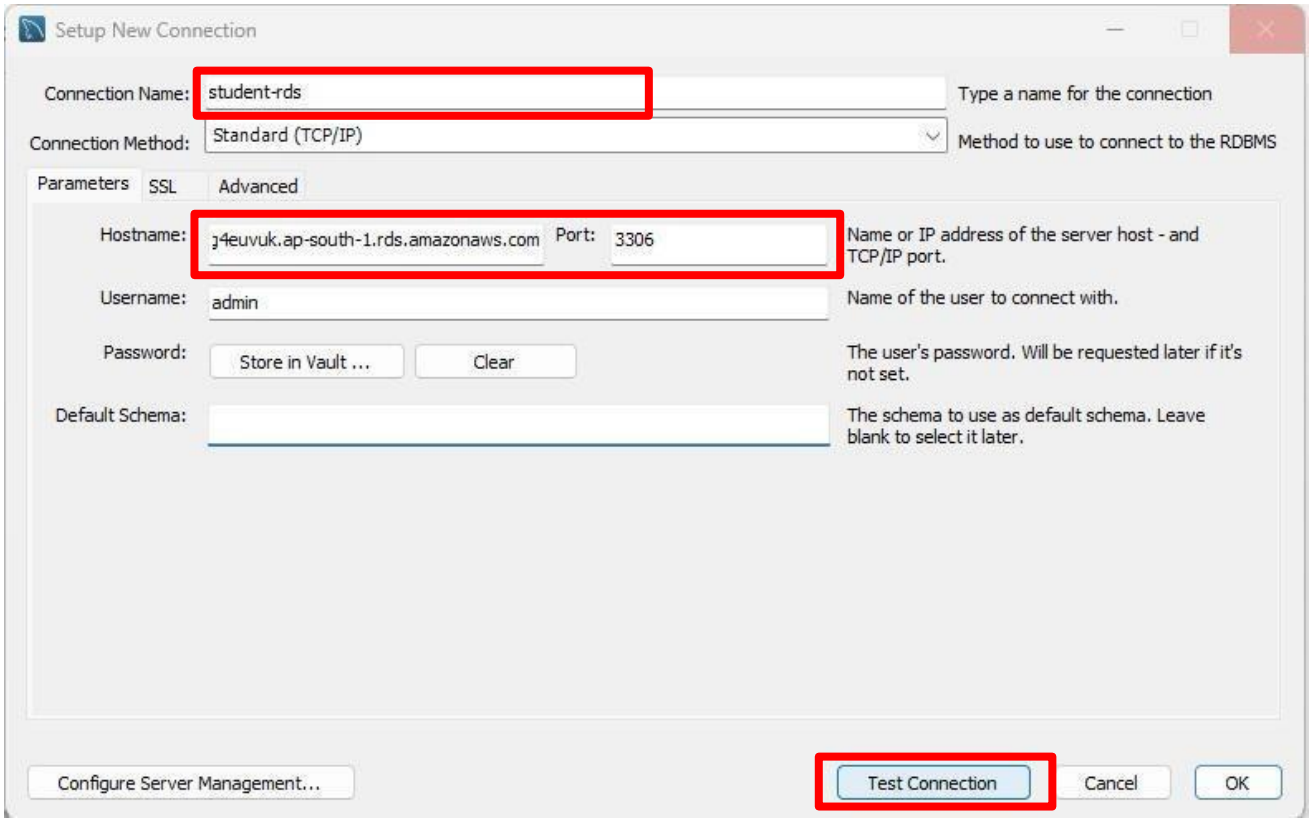
Host: RDS Endpoint

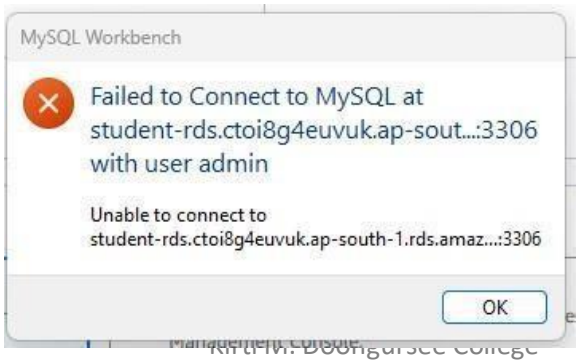
Username: admin

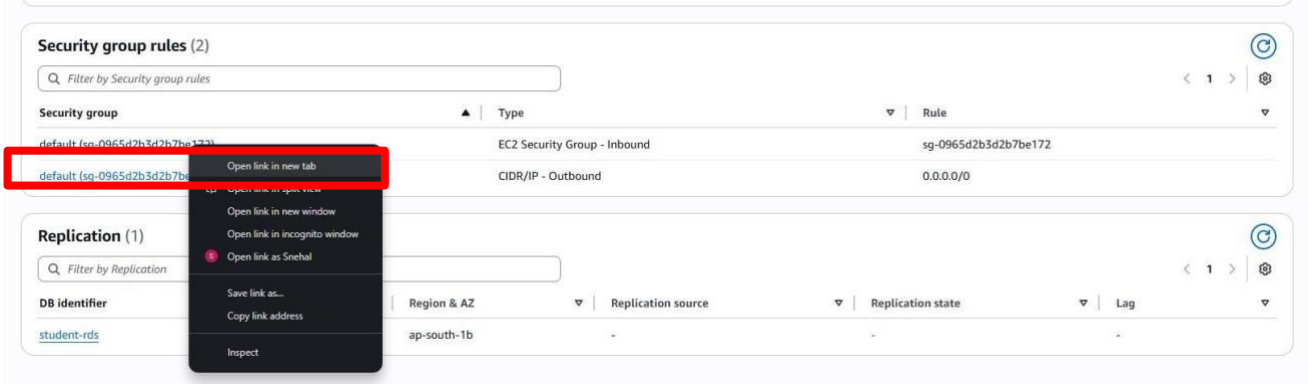
Password: (set earlier)

Port: 3306

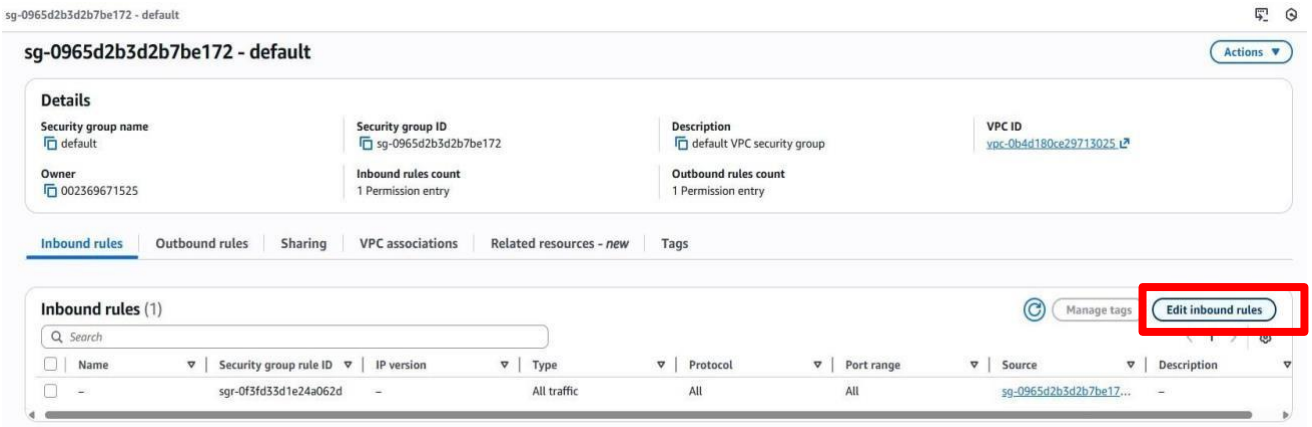

MySQL Workbench
File Edit View Database Tools Scripting Help
Welcome to MySQL Workbench
MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design,
create and browse y
design and run SQL quer
Browse Documentation
MySQL Connections
Local instance MySQL80
root
localhost:3306
Setup New Connection
Connection Name:
Type a name for the connection
Connection Method:
Standard (TCP/IP)
Method to use to connect to the RDBMS
Parameters SSL Advanced
Hostname:
127.0.0.1
Port:
3306
Name or IP address of the server host - and TCP/IP port.
Username:
root
Name of the user to connect with.
Password:
Store in Vault ...
Clear
The user's password. Will be requested later if it's not set.
Default Schema:
The schema to use as default schema. Leave blank to select it later.
Configure Server Management...
Test Connection
Cancel
OK


Setup New Connection
Connection Name:
student-rds
Type a name for the connection
Connection Method:
Standard (TCP/IP)
Method to use to connect to the RDBMS
Parameters SSL Advanced
Hostname:
g4euvuk.ap-south-1.rds.amazonaws.com
Port:
3306
Name or IP address of the server host - and TCP/IP port.
Username:
admin
Name of the user to connect with.
Password:
Store in Vault ...
Clear
The user's password. Will be requested later if it's not set.
Default Schema:
The schema to use as default schema. Leave blank to select it later.
Configure Server Management...
Test Connection
Cancel
OK

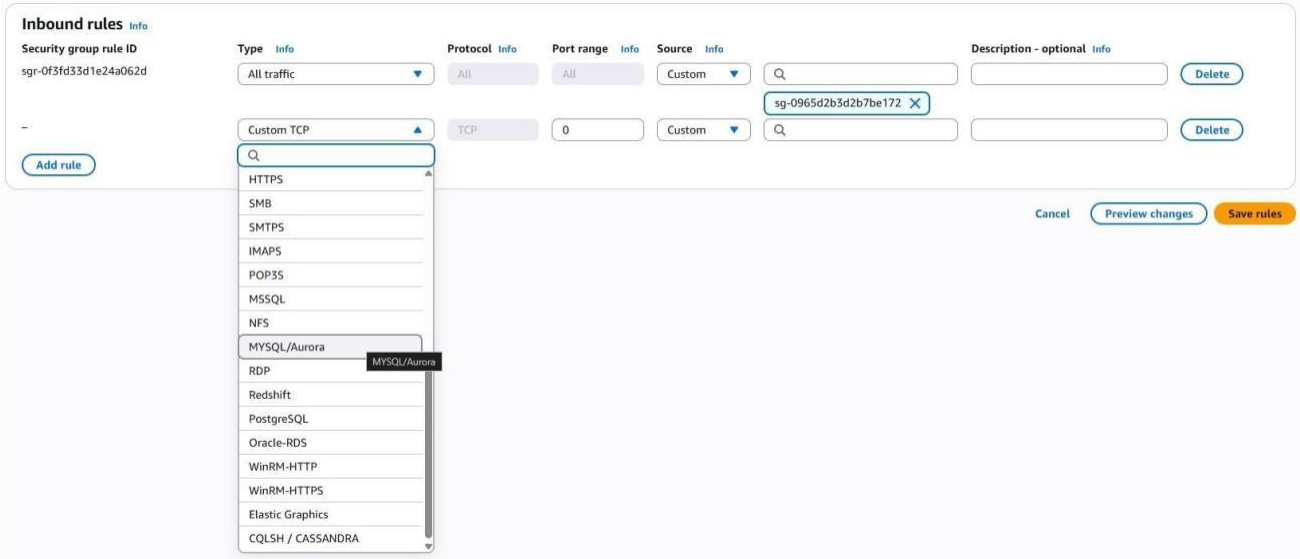

MySQL Workbench
Failed to Connect to MySQL at
student-rds.ctoi8g4euvuk.ap-sout...:3306
with user admin
Unable to connect to
student-rds.ctoi8g4euvuk.ap-south-1.rds.amaz...:3306
OK

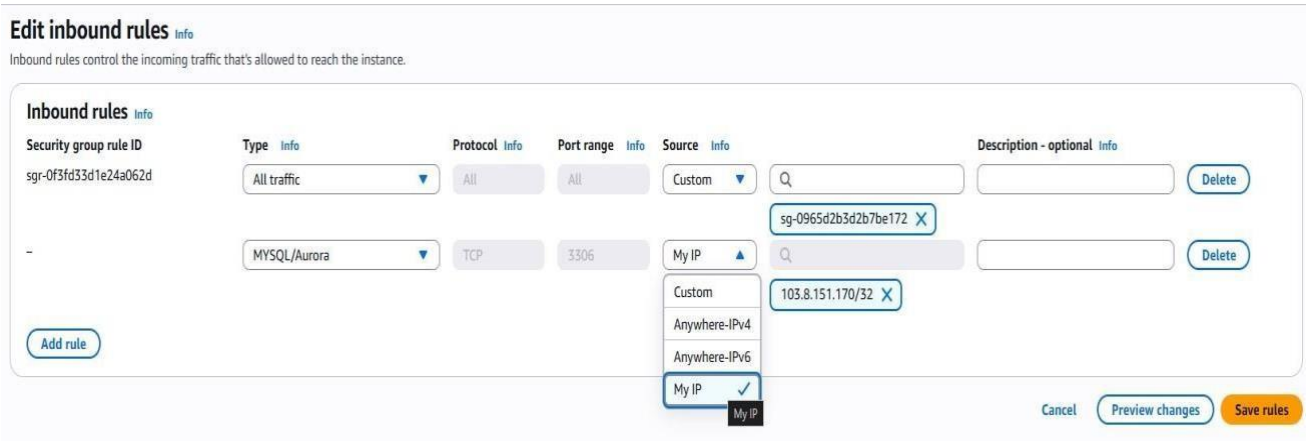

Security group rules (2)
Filter by Security group rules
Security group
Type
Rule
default (sg-0965d2b3d2b7be172)
EC2 Security Group - Inbound
sg-0965d2b3d2b7be172
default (sg-0965d2b3d2b7be
CIDR/IP - Outbound
0.0.0.0/0
Open link in new tab
Open link in new window
Open link in incognito window
Open link as Snehal
Save link as...
Copy link address
Inspect
Replication (1)
Filter by Replication
DB identifier
Region & AZ
Replication source
Replication state
Lag
student-rds
ap-south-1b

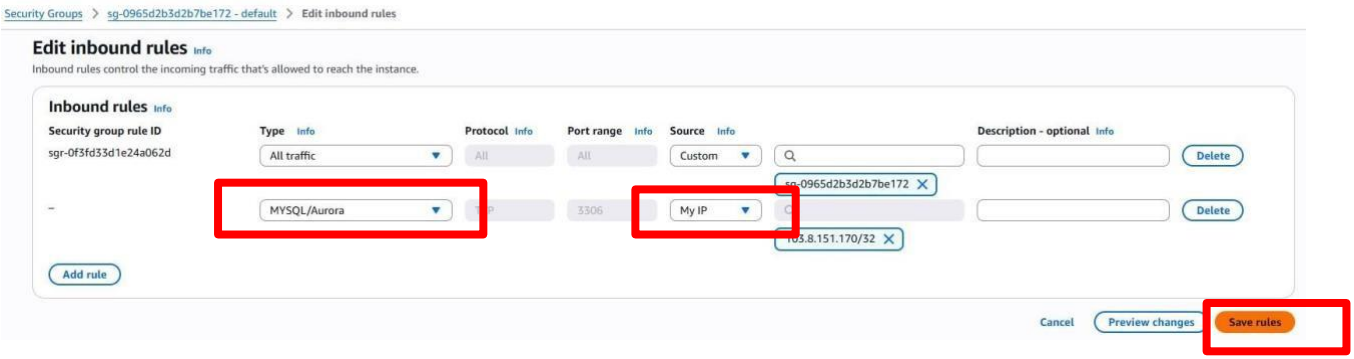
Security Groups (1) Info
Actions
Export security groups to CSV
Create security group
Find security groups by attribute or tag
sg-0965d2b3d2b7be172
Clear filters
Name
Security group ID
Security group name
VPC ID
Description
Owner
-
sg-0965d2b3d2b7be172
default
vpc-0b4d180ce29713025
default VPC security group
002369671525

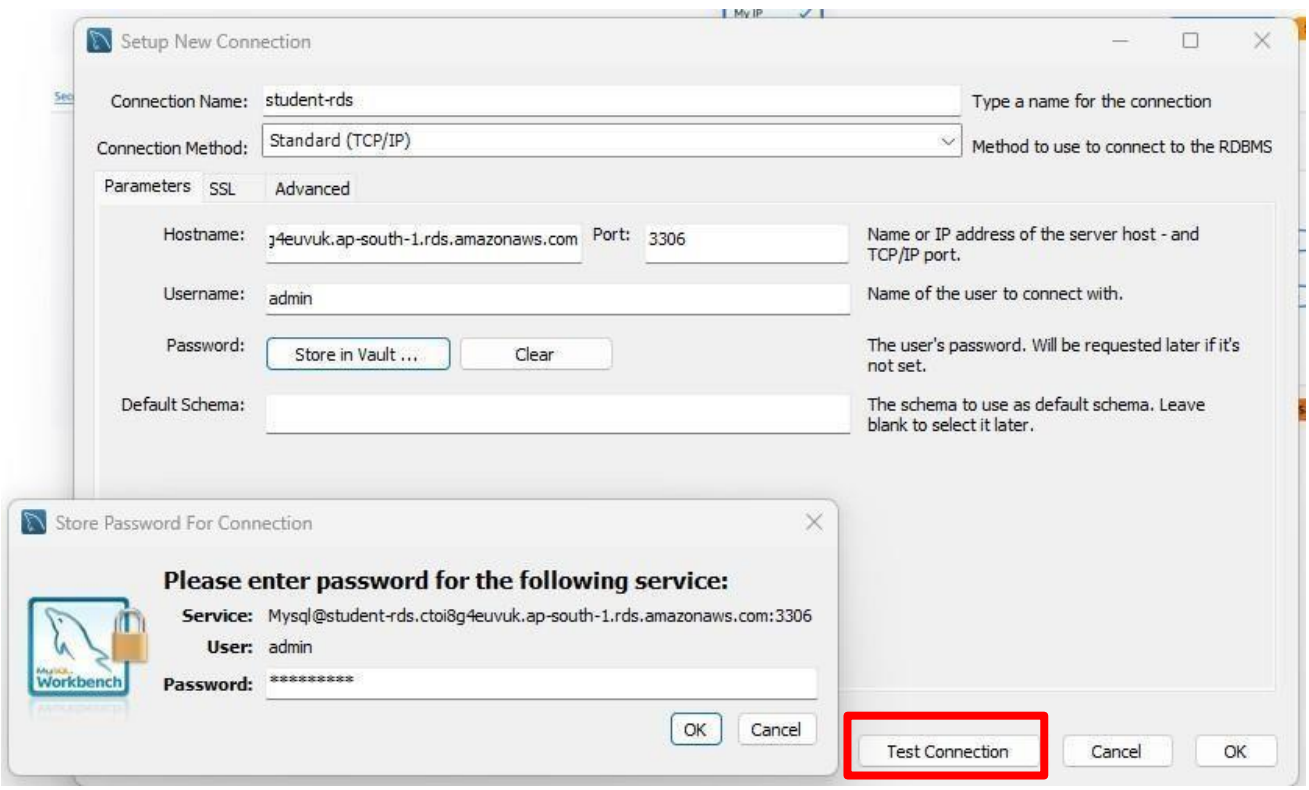

sg-0965d2b3d2b7be172 - default
sg-0965d2b3d2b7be172 - default
Actions
Details
Security group name
default
Security group ID
sg-0965d2b3d2b7be172
Description
default VPC security group
VPC ID
vpc-0b4d180ce29713025
Owner
002369671525
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
-
sgr-0f3fd33d1e24a062d
-
All traffic
All
All
sg-0965d2b3d2b7be17...
-

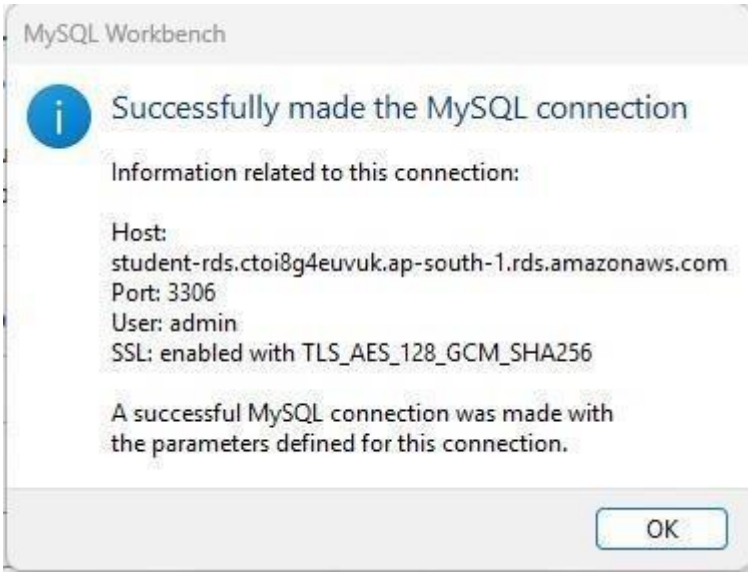
EC2 > Security Groups > sg-0965d2b3d2b7be172 - default > Edit inbound rules
Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
sgr-0f3fd33d1e24a062d
All traffic
All
All
Custom
sg-0965d2b3d2b7be172
Delete
Add rule
Cancel
Preview changes
Save rules

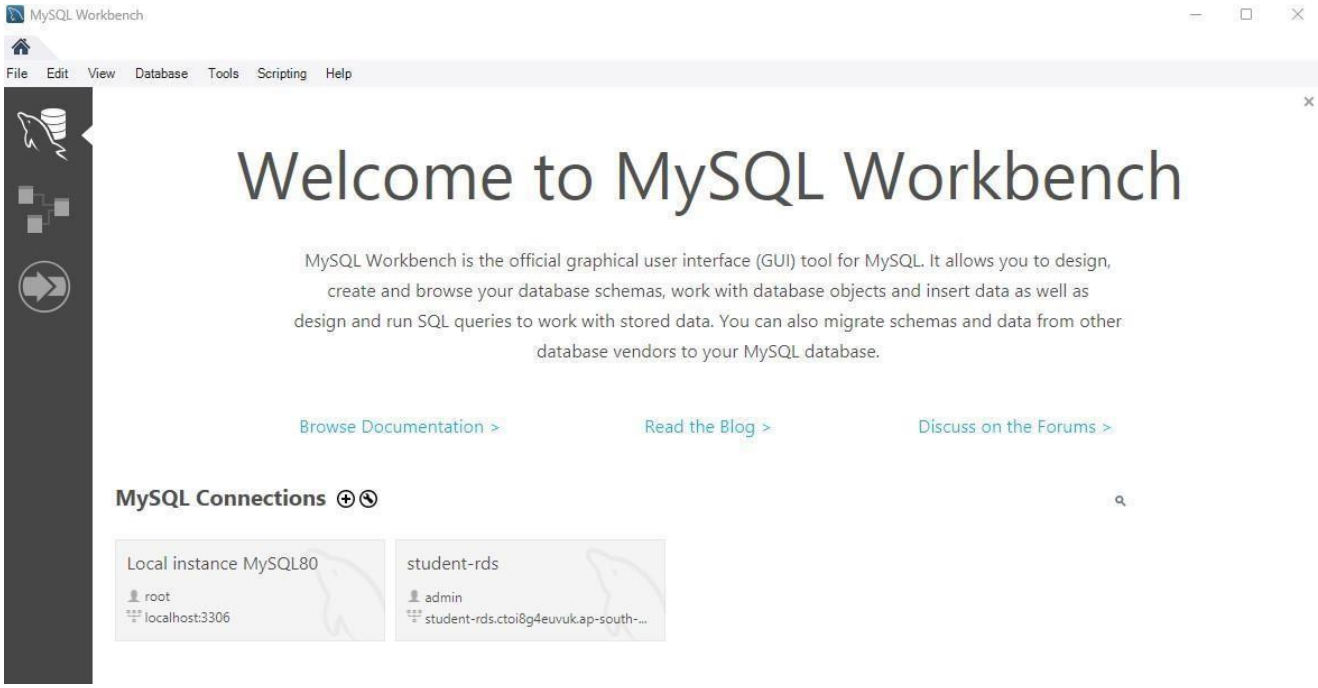

Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
sgr-0f3fd33d1e24a062d
All traffic
All
All
Custom
sg-0965d2b3d2b7be172
Delete
Custom TCP
TCP
0
Custom
Delete
Add rule
HTTPS
SMB
SMTPS
IMAPS
POP3S
MSSQL
NFS
MYSQL/Aurora
RDP
Redshift
PostgreSQL
Oracle-RDS
WinRM-HTTP
WinRM-HTTPS
Elastic Graphics
CQLSH / CASSANDRA
Cancel
Preview changes
Save rules

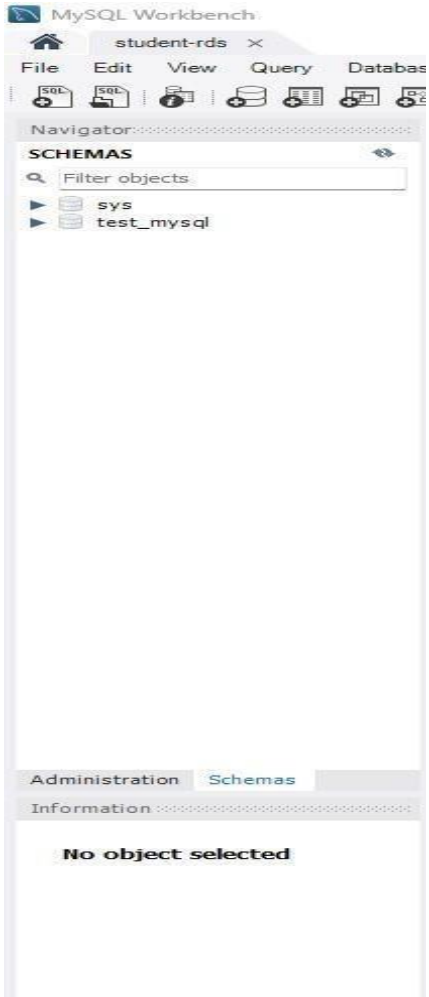

Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
sgr-0f3fd33d1e24a062d
All traffic
All
All
Custom
sg-0965d2b3d2b7be172
Delete
MYSQL/Aurora
TCP
3306
My IP
103.8.151.170/32
Delete
Custom
Anywhere-IPv4
Anywhere-IPv6
My IP
Add rule
Cancel
Preview changes
Save rules


Security Groups > sg-0965d2b3d2b7be172 - default > Edit inbound rules
Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
sgr-0f3fd33d1e24a062d
All traffic
All
All
Custom
sg-0965d2b3d2b7be172
Delete
MYSQL/Aurora
3306
My IP
103.8.151.170/32
Delete
Add rule
Cancel
Preview changes
Save rules


Setup New Connection
Connection Name: student-rds
Type a name for the connection
Connection Method: Standard (TCP/IP)
Method to use to connect to the RDBMS
Parameters SSL Advanced
Hostname: g4euvuk.ap-south-1.rds.amazonaws.com
Port: 3306
Name or IP address of the server host - and TCP/IP port.
Username: admin
Name of the user to connect with.
Password: Store in Vault ... Clear
The user's password. Will be requested later if it's not set.
Default Schema:
The schema to use as default schema. Leave blank to select it later.
Store Password For Connection
Please enter password for the following service:
Service: Mysql@student-rds.ctoi8g4euvuk.ap-south-1.rds.amazonaws.com:3306
User: admin
Password: *********
OK
Cancel
Test Connection
Cancel
OK


MySQL Workbench
Successfully made the MySQL connection
Information related to this connection:
Host:
student-rds.ctoi8g4euvuk.ap-south-1.rds.amazonaws.com
Port: 3306
User: admin
SSL: enabled with TLS_AES_128_GCM_SHA256
A successful MySQL connection was made with the parameters defined for this connection.
OK





Step 7: Perform Database Operations

CREATE DATABASE college;

USE college;

CREATE TABLE students (id INT

PRIMARY KEY AUTO_INCREMENT, name

VARCHAR(50), department

VARCHAR(30),

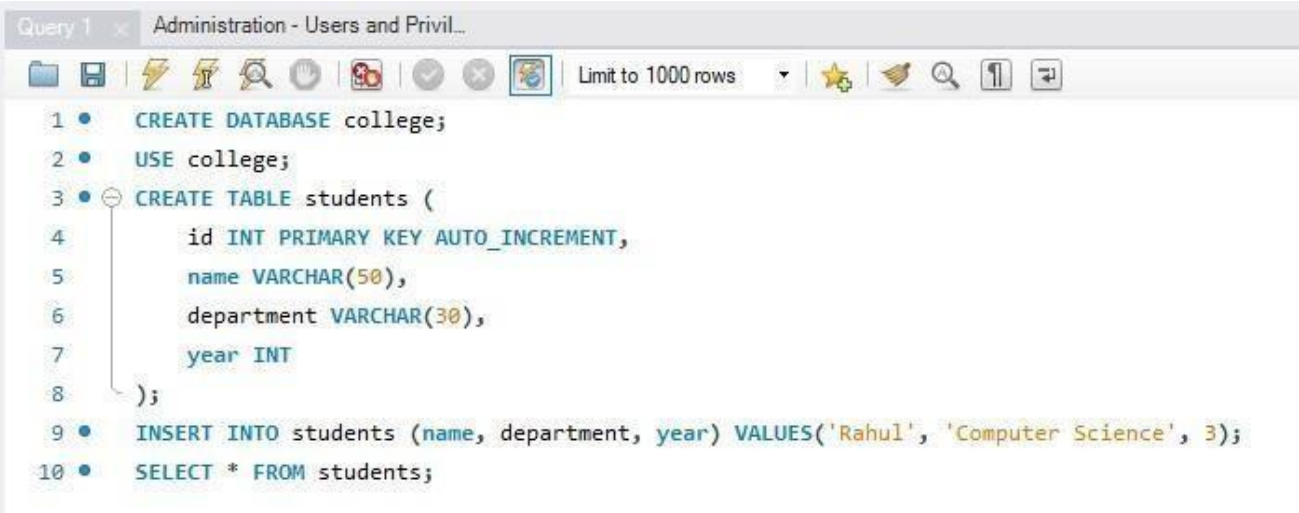
  year INT

);

INSERT INTO students (name, department, year)

VALUES ('Rahul', 'Computer Science', 3);

SELECT * FROM students;



Query 1 | Administration - Users and Privil...

Limit to 1000 rows

Execute the selected portion of the script or everything, if there is no selection

```
CREATE DATABASE college;
USE
CREATE TABLE students (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    department VARCHAR(30),
    year INT
);
INSERT INTO students (name, department, year) VALUES('Rahul', 'Computer Science', 3);
SELECT * FROM students;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name	department	year
1	Rahul	Computer Science	3
NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats | Execution Plan

students 1 | Apply | Revert

Output

Action Output

#	Time	Action	Message
1	13:49:39	CREATE DATABASE college	1 row(s) affected
2	13:49:39	USE college	0 row(s) affected
3	13:49:39	CREATE TABLE students (id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(50), department VARCHAR(30), year INT)	0 row(s) affected
4	13:49:39	INSERT INTO students (name, department, year) VALUES('Rahul', 'Computer Science', 3)	1 row(s) affected
5	13:49:39	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned

student-rds

student-rds

Modify | Actions

- Stop temporarily
- Reboot
- Delete
- Set up EC2 connection
- Set up Lambda connection
- Migrate data from EC2 database
- Create read replica
- Create Aurora read replica
- Create blue/green deployment
- Promote
- Convert to Multi-AZ deployment
- Take snapshot
- Restore to point in time
- Migrate snapshot
- Create zero-ETL integration
- Create ElastiCache cluster

Summary

DB identifier	Status	Role	Engine
student-rds	Available	Instance	MySQL Community
CPU	**Class**	**Current activity**	**Region & AZ**
-	db.t4g.micro		ap-south-1b

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations | Tags

Connect using Info

- Code snippets — Use when connecting through SDK, APIs, or third-party tools including agents.
- CloudShell — Use for a quick access to AWS CLI that launches directly from the AWS Management Console.
- Endpoints — Use when connecting through any IDE in

Internet access gateway: Disabled

IAM Authentication: Disabled

Programming language: MySQL (Windows)

Endpoint type: Instance endpoint

Connection steps

Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

PowerShell

```
$pwd = <Enter_DB_Password>
mysql -h student-rds.cto18g4euvuk.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p"$pwd" --ssl-verify-server-cert  --ssl-ca="C:\certs\global-bundle.pem" test_mysql
```

Output

AWS RDS MySQL instance created successfully

Database and table created

Data inserted and retrieved from RDS